

Stormwater Management as a Sociotechnical System: A Comparison of Virginia and North Carolina

A Research Paper submitted to the Department of Engineering and Society

Presented to the Faculty of the School of Engineering and Applied Science
University of Virginia • Charlottesville, Virginia

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science, School of Engineering

Chelsey Ojeda – San Juan

Spring 2026

On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

Advisor

Pedro A. P. Francisco, Department of Engineering and Society

Introduction:

With the increase in urbanization, land development becomes more impervious and disturbed from what it was previously, which was forested or vegetated land. With more land being developed, impervious surfaces such as rooftops, parking lots, and sidewalks replace natural ground cover. Due to this, stormwater behaves differently, and its behavior is significantly altered. Rainwater moves differently when it encounters impervious surfaces, as it can no longer infiltrate into natural ground cover, therefore increasing the velocity of the runoff. This runoff then carries sediment and pollutants into a nearby storm drain, set to outfall into nearby water bodies. The repetition of this process causes stream channel deterioration, increased erosion, and negatively impacts water quality over time. Therefore, it is important to mitigate these impacts through stormwater infrastructure and design (U.S. Environmental Protection Agency, n.d.-a).

However, the responsibility does not just fall on the engineer who is in charge of designing a stormwater infrastructure system. Stormwater is regulated through policy and involves multiple actors, including developers, government agencies, engineers, and local communities. Due to this, stormwater infrastructure, design, and policy can be understood as a sociotechnical system. A sociotechnical system means that there is not only a physical system in place, but it is also shaped by policy, regulation, and human decision-making (Chris Clegg et al., 2000). Infrastructure design is based on environmental conditions, economic considerations, and social priorities. Engineers, for example, do not design freely; they are given many constraints for their design. Some of these constraints are defined by permits, state regulations, and federal requirements. However, they also have to balance cost, land use compliance, and client

preferences for each of their projects, while complying with and enforcing environmental protection goals. Stormwater is managed by multiple actors contributing to the process.

Some of the main concerns following urbanization are flooding and water quality issues. In an undisturbed environment, when it rains, water has the opportunity to slow down and infiltrate into the groundwater system before reaching nearby water bodies. However, in a developed area, impervious surfaces cause water to rush off the surface. These surfaces are typically exposed to a lot of debris and pollutants. These pollutants include nutrients, sediments, heavy metals, bacteria, oil, cleaning supplies, etc. Therefore, all of this is washed off with the rain and is transferred directly into streams and rivers (U.S. Environmental Protection Agency, n.d.-a). Additionally, the natural slowing and spreading of water is lost. The natural spread and infiltration of water reduce the amount of runoff entering waterways at a given point in time. Without this process, water moves faster and is carried in a much larger volume. This increases the risk of flooding, channel erosion, and strain on stormwater infrastructure (U.S. Environmental Protection Agency, n.d.-a).

In addition to water quality impacts, stormwater drastically affects ecosystem health and long-term watershed stability. An increase in runoff can significantly alter a stream or channel's natural structure, degrade aquatic habitats, and disrupt biological communities. The transport of sediment can inhibit the access of aquatic life to their habitats. Additionally, nutrient loading can lead to eutrophication and algal blooms. These impacts further demonstrate that stormwater is not just an infrastructure issue, but an environmental system that requires careful management and responsibility (U.S. Environmental Protection Agency, n.d.-b).

Another environmental aspect that directly impacts stormwater infrastructure is climate change. Recently, climate change has been a growing concern, and with an increase in awareness, it is important to consider how stormwater infrastructure systems will need to adapt. With an increase in temperatures comes more intense and frequent storm events, which place additional strain on stormwater systems (U.S. Environmental Protection Agency, 2017). Many of these systems have been built and designed using historical rainfall data, which do not consider long-term impacts. This creates uncertainty and concern about existing stormwater systems and future conditions (Chesapeake Bay Program, 2020). Therefore, stormwater systems need to be designed with adaptability and resilience in mind. A poorly built stormwater system can have detrimental impacts on local communities, such as flooding, which also threatens public safety. Additionally, it threatens environmental health. Therefore, engineers will design according to what they are required to follow by the state; thus, the majority of the responsibility falls on the policies that are implemented.

Stormwater is managed through a network of actors, from federal to state agencies, to local governments and residents, and engineers working with developers' expectations. In the United States, federal policies such as the Clean Water Act establish requirements for water quality (U.S. Environmental Protection Agency, 2023). However, each state is given the liberty to establish its own process for stormwater management through state regulation and local ordinances. Meaning that each state follows the same federal framework; however, the policies that each puts in place can vary significantly. Therefore, for this study, stormwater policy and water quality outcomes will be examined in Virginia and North Carolina. These states were selected due to similar climate, geography, and patterns of development. However, they differ in how they manage stormwater and how it is implemented at the state level. The study asks how

differences in stormwater policy between Virginia and North Carolina shape infrastructure design, water quality outcomes, and regulatory implementation (U.S. Environmental Protection Agency, 2023).

Ultimately, the study finds that stormwater policy influences both technical systems and social structures. Virginia's stormwater system is more centralized, which also facilitates tracking water quality improvements over time. North Carolina, on the other hand, uses a more decentralized, site-specific system for stormwater management. This results in fewer comparable statewide outcomes. This study demonstrates the connection between infrastructure policy and environmental outcomes (Clegg et al., 2000; Virginia Department of Environmental Quality, n.d.; U.S. Environmental Protection Agency, 2023).

Background:

Both Virginia and North Carolina are federally governed and constrained by the Clean Water Act (CWA), which regulates water pollution in the United States. Additionally, the CWA establishes the legal framework that states must meet for controlling pollutant discharges and protecting water quality (U.S. Environmental Protection Agency, n.d.-b). One regulatory program under the CWA is the National Pollutant Discharge Elimination System (NPDES). The NPDES program requires permits for point source discharges into surface waters and sets limits, monitoring, and reporting requirements to control what is released into water bodies (U.S. Environmental Protection Agency, n.d.-a). These permits enforce requirements for each discharge location in order to comply with the Clean Water Act (U.S. Environmental Protection Agency, n.d.-a).

Stormwater discharges from Municipal Separate Storm Sewer Systems (MS4s) are a major component of NPDES regulation. MS4 systems are publicly owned stormwater infrastructure networks, including drains, pipes, and outlets, which discharge untreated runoff directly into water bodies. Essentially, because these systems do not treat water that is discharged, they contribute significantly to non-point source pollution (U.S. Environmental Protection Agency, n.d.-c). A point source of pollution is any infrastructure system, such as a pipe or ditch, in which a pollutant is directly discharged into a water body.

MS4 systems are regulated at the state level through programs such as the Virginia Pollutant Discharge Elimination System (VPDES), which implements federal NPDES requirements within Virginia (Virginia Department of Environmental Quality, n.d.-a). Therefore, within MS4 permits, it is a requirement that operators implement control measures for pollutant discharge, have public education programs, construction site runoff controls, and maintenance practices (Virginia Department of Environmental Quality, n.d.-d). In North Carolina, its own NPDES permitting system is implemented through the North Carolina Department of Environmental Quality, which enforces state-level water quality requirements under the federal Clean Water Act framework.

These systems are critical because they directly influence how engineers design stormwater systems. Engineers are held to these requirements, and due to client pressure, many will try to design as close to meeting minimum requirements as possible. This is mainly driven by cost constraints and the pressure of completing projects within budget and schedule. For example, permitting requirements may include allowable discharge limits, required treatment practices, and construction and operational compliance standards (North Carolina Department of Environmental Quality, n.d.-a).

In Virginia, stormwater management is influenced by the Chesapeake Bay watershed. The Chesapeake Bay experiences significant nutrient pollution, with the main sources being nitrogen and phosphorus. This has led to widespread environmental impacts. Whenever there is an excess of nutrients within a water body, algal blooms become a major risk. Algal blooms reduce oxygen levels, which harm aquatic life and overall ecosystem health (U.S. Environmental Protection Agency, n.d.-a). Therefore, to help with this effort, the Chesapeake Bay Total Maximum Daily Load (TMDL) was set in place. The TMDL sets limits on the amount of pollutants that can enter the watershed and requires the state to implement strategies to meet those limits (Chesapeake Bay Program, n.d.). The TMDL has led Virginia to adopt a more centralized and structured approach to stormwater management. One thing that developers must submit with their projects, if necessary, is a detailed stormwater management plan that includes best management practices, such as detention ponds, infiltration, and green infrastructure (Virginia Department of Environmental Quality, n.d.-b). This practice is meant to reduce runoff volume, improve water quality, and mimic natural stormwater processes.

In North Carolina, a more decentralized approach is used for stormwater management. Stormwater management is mainly administered through the North Carolina Department of Environmental Quality (NCDEQ). However, the focus shifts to post-construction stormwater programs and watershed-specific nutrient strategies. The post-construction program regulates stormwater after development has been completed, requiring BMP implementation in certain regulated areas (North Carolina Department of Environmental Quality, n.d.-b). North Carolina also relies on basin-specific nutrient strategies. These strategies are specific to individual watersheds, such as the Tar-Pamlico River Basin and Jordan Lake. Nutrient reduction, impervious surface limits, and stormwater controls are required under these strategies (North

Carolina Department of Environmental Quality, n.d.-c; North Carolina Department of Environmental Quality, n.d.-d). This approach allows policies to be more specific according to regions and their needs, but it does result in more variability across the state.

Therefore, this difference in Virginia and North Carolina policy structure tells us that policy plays a major role in stormwater infrastructure, how it is designed, and how outcomes will ultimately be measured.

Methodology:

The goal of this case study between Virginia and North Carolina is to analyze stormwater policies and water quality results. Primary sources include integrated water quality reports, stormwater program documentation, NCDEQ stormwater policy outlines, and VADEQ stormwater policy outlines, along with nutrient management strategies (Virginia Department of Environmental Quality, n.d.-c; North Carolina Department of Environmental Quality, n.d.-e). These sources focus on quantitative results for water quality, along with policy context and frameworks. Secondary analysis focuses on interpreting these policies and how they could impact or correlate with water quality results.

Analysis focuses on several key areas, such as regulatory structure and nutrient management strategies, which ultimately influence developer compliance requirements. These areas are important and were selected because they impact stormwater infrastructure systems, how they are designed, and how effective they are. In addition to these technical aspects, the study evaluates sociotechnical outcomes. This includes how responsibility is distributed among agencies and stakeholders, how regulatory requirements impact engineering decisions, and how

monitoring systems shape our understanding of environmental conditions and improvements (Clegg et al., 2000).

Actor Network Theory (ANT) is a sociotechnical framework that examines how society is formed through networks of human and non-human actors interacting with one another. Instead of treating social and technical systems as separate, ANT views them as part of the same network, where different actors play a role in shaping outcomes (Latour, 2005). In this case study, Actor Network Theory helps show the relationships between actors such as the EPA, state agencies, engineers, developers, and infrastructure systems. This includes analyzing how federal policies are broken down and translated differently from state to state, and how those policies become integrated into design constraints through permits that ultimately shape infrastructure decisions and water quality outcomes.

An additional framework that is important for this case study is Responsible Research and Innovation (RRI). RRI focuses on how research and innovation processes can meet societal needs by ensuring they anticipate future impacts within current processes, rather than being reactive after impacts have already occurred (Stilgoe, Owen, & Macnaghten, 2013). Therefore, it is important to evaluate whether stormwater policies are proactive or reactive, and whether they account for long-term environmental changes such as climate variability or increased urbanization. This framework also shows how responsibility is distributed among the different actors identified in Actor Network Theory. Additionally, it helps assess whether stormwater management systems in Virginia and North Carolina allow for continuous improvement through monitoring and adaptation.

However, there are limitations, such as the fact that monitoring methods differ between Virginia and North Carolina, making direct comparisons less consistent. Additionally, environmental

improvement is a long-term process, meaning noticeable differences require extended timeframes. Despite these limitations, the study still provides an overview of how policy and infrastructure interact across both states.

Discussion:

The research question seeks to determine whether there are measurable improvements in water quality in Virginia and North Carolina, and what differences in monitoring and policy reveal about those outcomes. Both states monitor statewide water quality, but there are significant differences, similar to the differences seen in stormwater management policy. However, due to these differences in monitoring, direct comparison of water quality between Virginia and North Carolina is difficult. Virginia's system shows clearer trends and improvement over time, while North Carolina focuses more on identifying currently impaired waters. This difference highlights how monitoring, policy, and data collection shape how environmental progress and improvements are understood.

Both states provide Integrated Reports under Sections 305(b) and 303(d) of the Clean Water Act. In Virginia, water bodies are assessed every two years across six areas: aquatic life, fish consumption, public water supply, recreation, shellfishing, and wildlife (Virginia Department of Environmental Quality, n.d.-c). If one of these areas does not meet the requirements, the water body is classified as impaired and requires a restoration plan.

Looking at the 2020 and 2022 reports for Virginia, it shows clear and measurable trends. For example, river impairments remained stable at 16%, while non-impaired rivers declined slightly from 6% to 4%. Lakes increased in impairment from 80% to 86%, while non-impaired lakes decreased from 16% to 10%. Estuarine impairments remained constant at 75%. These

assessments were conducted over 100,983 miles of rivers, 2,843 square miles of estuarine waters, and 117,753 acres of lakes and reservoirs (Virginia Department of Environmental Quality, 2022). While some increases in impairment could be influenced by new data and improved monitoring, the ability to track percentages over time provides a clear, quantifiable way to evaluate improvements or impairments.

The causes of impairment also highlight the role of stormwater management. Bacteria affect 12,346 miles of rivers, toxins in fish tissue affect 3,702 miles, and low dissolved oxygen impacts 704 miles (Virginia Department of Environmental Quality, 2022). In lakes, toxics affect 87,849 acres, and dissolved oxygen impairment affects 47,590 acres. These pollutants are traced back directly to stormwater runoff, which transports nutrients, bacteria, and heavy metals into water bodies. Because Virginia continuously tracks these values, it provides a measurable way to see how policies such as the Chesapeake Bay TMDL are influencing water quality outcomes over time.

On the other hand, North Carolina's 2022 Integrated Report and Water Management Plans demonstrate a different approach. While assessments are conducted regularly, impairment is determined using a five-year dataset. This means that water quality is evaluated based on long-term exceedance of standards rather than short-term trends (North Carolina Department of Environmental Quality, n.d.-e). This approach focuses more on classification rather than tracking improvement over time. Additionally, North Carolina relies heavily on basin-specific monitoring rather than statewide summaries. One example is that Jordan Lake and Falls Lake have been identified as impaired waters due to nutrient-related issues, including chlorophyll-a concentrations greater than 40 $\mu\text{g/L}$, turbidity greater than 25 NTU, dissolved oxygen levels below 4 mg/L, and pH outside the range of 6–9 (North Carolina Department of Environmental

Quality, 2020). These thresholds define when water quality standards are exceeded, rather than measuring gradual improvement.

Additionally, monitoring in these lakes is based on approximately 11 sampling events per year at multiple stations. Results are summarized using averages, minimums, maximums, and percent exceedance of standards (North Carolina Department of Environmental Quality, 2020). While this produces very detailed datasets at the local level, it does not translate into a clear, statewide, traceable measurement approach. Because of this, North Carolina shows strong technical data within individual watersheds, but instead of focusing on water quality improvement over time, it mainly shows whether standards are being exceeded within specific basins. Therefore, North Carolina focuses more on identifying impairment rather than demonstrating measurable statewide improvement.

Using Actor-Network Theory (ANT), the differences between Virginia and North Carolina's stormwater and water quality governance systems become clearer. In Virginia, the network of actors—including the U.S. Environmental Protection Agency (EPA), state agencies such as the Virginia Department of Environmental Quality, the Chesapeake Bay Program, scientific monitoring systems, and TMDL (Total Maximum Daily Load) requirements—is highly interconnected. Chesapeake Bay Program requirements strongly shape state policy, which in turn drives continuous monitoring and adaptive engineering responses. Because water quality data is collected and updated regularly, a feedback loop is created in which science, policy, and infrastructure continuously inform one another (Latour, 2005; Chesapeake Bay Program, n.d). This tightly coupled network enables Virginia to track measurable trends over time and evaluate environmental improvements in a structured, quantifiable way.

In contrast, North Carolina's governance network is more decentralized. Key actors include the North Carolina Department of Environmental Quality, basin-specific management programs such as those for Jordan Lake and Falls Lake, and localized monitoring efforts. While these systems also operate under federal Clean Water Act requirements, they function more independently at the watershed scale, with management decisions largely confined within basin boundaries. From an ANT perspective, this produces weaker statewide connectivity between monitoring, policy adjustment, and infrastructure investment. In many cases, data is primarily used to determine impairment status rather than to continuously track long-term improvement trends. As a result, while highly detailed local data exists, it is more difficult to identify consistent statewide environmental trajectories.

When analyzed through the lens of Responsible Research and Innovation (RRI), these differences highlight contrasting approaches to environmental governance. RRI emphasizes anticipation, responsiveness, and the integration of knowledge into long-term sustainable decision-making (Stilgoe, Owen, & Macnaghten, 2013). Virginia's system aligns more closely with these principles due to its continuous monitoring framework and TMDL-driven feedback mechanisms, which allow for earlier intervention and adaptive management before conditions significantly deteriorate. This enables the state to adjust strategies based on measurable pollutant trends, reflecting a more proactive governance model.

North Carolina's system, by comparison, is more reactive. Although it is effective at identifying when water quality standards are exceeded, regulatory action is often initiated after impairment has already been documented, frequently based on multi-year assessment cycles. However, its basin-specific structure does provide flexibility for localized responses tailored to individual watershed conditions, which still aligns with aspects of RRI in terms of contextualized

decision-making. The limitation lies less in data quality and more in the lack of an integrated statewide mechanism for tracking long-term progress.

Overall, Virginia's more centralized and continuously monitored system allows for clearer evaluation of water quality improvements over time, while North Carolina's decentralized, basin-specific approach produces detailed local insights but limits statewide comparability. Both systems operate under the same federal regulatory framework, yet differences in monitoring structure and policy connectivity significantly influence how environmental outcomes are measured, interpreted, and acted upon.

Conclusion:

The study set out to determine whether measurable improvements in water quality can be identified in Virginia and North Carolina, and how differences in monitoring and policy frameworks shape those outcomes. The results indicate that Virginia's system demonstrates clear, quantifiable trends in water quality improvement over time, while North Carolina's system provides more detailed but less directly comparable datasets at a statewide level. Through Actor-Network Theory, these differences can be understood as a product of how each state structures the relationships between policy, monitoring systems, and engineering design. In Virginia, the more centralized system creates a tightly connected network in which monitoring data directly informs policy decisions and infrastructure interventions. In contrast, North Carolina's decentralized, basin-based structure produces highly detailed localized water quality reports; however, this fragmentation may reduce the ability for findings to be synthesized into broader statewide trends or shared effectively across watershed boundaries.

From a Responsible Research and Innovation (RRI) perspective, it is essential to recognize that environmental decision-making is intended to anticipate future conditions rather than only respond to existing impairment. Virginia demonstrates a more anticipatory approach by using continuous monitoring systems and nutrient loading limits to guide long-term water quality improvements. This enables policy and engineering decisions to be adjusted proactively based on observed trends. North Carolina, however, operates in a more reactive manner, primarily identifying issues once water quality standards have already been exceeded rather than continuously tracking progress toward improvement targets. While this approach still supports regulatory compliance, it places greater emphasis on correction after impairment rather than prevention.

Future research should focus on developing standardized methods for comparing water quality outcomes across different monitoring and governance structures. This would improve the ability to evaluate environmental performance between states with fundamentally different policy frameworks. Additionally, further investigation is needed into how climate change and increasing urbanization may affect both centralized and decentralized stormwater management systems. Expanding this type of analysis to include additional states would also provide a more comprehensive understanding of how governance structures influence environmental outcomes at larger scales.

Overall, this study highlights that stormwater management is not solely a technical engineering challenge, but a socio-technical system shaped by policy, monitoring networks, and human decision-making. The key takeaway is that differences in governance and monitoring structures not only influence water quality outcomes but also determine how those outcomes are

measured, interpreted, and acted upon. Recognizing these differences is essential for developing more effective, adaptable, and equitable stormwater management strategies in the future.

References

- American Rivers. (2025, January 16). Keeping pollution out of your water: How the Clean Water Act protects your water. <https://www.americanrivers.org/2025/01/keeping-pollution-out-of-your-water-how-the-clean-water-act-protects-your-water/>
- Chesapeake Bay Program. (2020). Summary of stormwater design standards review. https://www.chesapeakebay.net/files/documents/memo_2_summary_of_stormwater_design_standards_review_draft_9.4.20.pdf
- Chesapeake Bay Program. (n.d.). Total Maximum Daily Load (TMDL). <https://www.chesapeakebay.net/what/programs/total-maximum-daily-load>
- Clegg, C. W. (2000). Sociotechnical principles for system design. *Applied Ergonomics*, 31(5), 463–477. [https://doi.org/10.1016/S0003-6870\(00\)00009-0](https://doi.org/10.1016/S0003-6870(00)00009-0)
- Latour, B. (2005). *Reassembling the social: An introduction to actor-network theory*. Oxford University Press. <https://www.uni-weimar.de/kunst-und-gestaltung/wiki/images/Latour-introduction-to-ant-theory.pdf>
- North Carolina Department of Environmental Quality. (2020). Jordan Lake monitoring report. <https://edocs.deq.nc.gov/WaterResources/DocView.aspx?dbid=0&id=2738822&cr=1>
- North Carolina Department of Environmental Quality. (n.d.-a). NPDES wastewater permitting. <https://www.deq.nc.gov/about/divisions/water-resources/water-quality-permitting/npdes-wastewater>

North Carolina Department of Environmental Quality. (n.d.-b). Post-construction stormwater program. <https://www.deq.nc.gov/about/divisions/energy-mineral-and-land-resources/stormwater/stormwater-program/post-construction-program>

North Carolina Department of Environmental Quality. (n.d.-c). Tar-Pamlico nutrient strategy. <https://www.deq.nc.gov/about/divisions/water-resources/water-planning/nonpoint-source-planning/tar-pamlico-nutrient-strategy>

North Carolina Department of Environmental Quality. (n.d.-d). Jordan Lake nutrient strategy. <https://www.deq.nc.gov/about/divisions/water-resources/water-planning/nonpoint-source-planning/jordan-lake-nutrient-strategy>

North Carolina Department of Environmental Quality. (n.d.-e). Water quality data assessment. <https://www.deq.nc.gov/about/divisions/water-resources/water-planning/modeling-assessment/water-quality-data-assessment>

North Carolina Department of Environmental Quality. (n.d.-f). Falls and Jordan Lakes monitoring. <https://www.deq.nc.gov/about/divisions/water-resources/water-sciences-section/intensive-survey-branch-isb/falls-and-jordan-lakes-monitoring>

Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. *Research Policy*, 42(9), 1568–1580. <https://doi.org/10.1016/j.respol.2013.05.008>

U.S. Environmental Protection Agency. (2017). Climate adaptation and stormwater runoff. https://19january2017snapshot.epa.gov/arc-x/climate-adaptation-and-stormwater-runoff_.html

U.S. Environmental Protection Agency. (n.d.-a). NPDES stormwater program.

<https://www.epa.gov/npdes/npdes-stormwater-program>

U.S. Environmental Protection Agency. (n.d.-b). Basic information about nutrient pollution.

<https://www.epa.gov/nutrientpollution/basic-information-nutrient-pollution>

U.S. Environmental Protection Agency. (n.d.-c). Stormwater discharges from municipal sources.

<https://www.epa.gov/npdes/stormwater-discharges-municipal-sources>

U.S. Environmental Protection Agency. (n.d.-d). NPDES permit basics.

<https://www.epa.gov/npdes/npdes-permit-basics>

Virginia Department of Environmental Quality. (n.d.-a). Virginia pollutant discharge elimination system (VPDES). [https://www.deq.virginia.gov/news-](https://www.deq.virginia.gov/news-info/shortcuts/permits/water/surface-waters-vpdes)

[info/shortcuts/permits/water/surface-waters-vpdes](https://www.deq.virginia.gov/news-info/shortcuts/permits/water/surface-waters-vpdes)

Virginia Department of Environmental Quality. (n.d.-b). Stormwater management.

<https://www.deq.virginia.gov/water/stormwater>

Virginia Department of Environmental Quality. (n.d.-c). Integrated report.

<https://www.deq.virginia.gov/water/water-quality/assessments/integrated-report>

Virginia Department of Environmental Quality. (n.d.-d). MS4 program.

<https://www.deq.virginia.gov/news-info/shortcuts/permits/water/ms4>